

6(a)(i)	<i>any one from:</i> 1 same / equal, size / length ; 2 same, shape / position of centromere ; 3 same banding pattern ;	1
6(a)(ii)	<i>any three from:</i> 1 recessive / mutation = faulty / inactive / no, tyrosinase ; 2 (melanocytes) do not produce, melanin / pigment ; 3 <i>ref. to</i> skin / hair / iris ; 4 AVP ;	3

Question	Answer	Marks																									
6(b)	<i>gametes (on Punnett square)</i> 1 BF Bf bF bf ; <i>Punnett square</i> <table><tr><td></td><td>BF</td><td>Bf</td><td>bF</td><td>bf</td></tr><tr><td>BF</td><td>BBFF black</td><td>BBFf black</td><td>BbFF black</td><td>BbFf black</td></tr><tr><td>Bf</td><td>BBFf black</td><td>BBff white</td><td>BbFf black</td><td>Bbff white</td></tr><tr><td>bF</td><td>BbFF black</td><td>BbFf black</td><td>bbFF brown</td><td>bbFf brown</td></tr><tr><td>bf</td><td>BbFf black</td><td>Bbff white</td><td>bbFf brown</td><td>bbff white</td></tr></table> 2 correct offspring genotypes on Punnett square ; 3 phenotypes matching genotypes ; <i>offspring phenotype ratio</i> 4 9 black : 3 brown : 4 white ;		BF	Bf	bF	bf	BF	BBFF black	BBFf black	BbFF black	BbFf black	Bf	BBFf black	BBff white	BbFf black	Bbff white	bF	BbFF black	BbFf black	bbFF brown	bbFf brown	bf	BbFf black	Bbff white	bbFf brown	bbff white	4
	BF	Bf	bF	bf																							
BF	BBFF black	BBFf black	BbFF black	BbFf black																							
Bf	BBFf black	BBff white	BbFf black	Bbff white																							
bF	BbFF black	BbFf black	bbFF brown	bbFf brown																							
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